Econ 201B: Lecture 1

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1 Uncertainty and Strategic Uncertainty

1.1 Uncertainty vs Strategic Uncertainty

Definition 1. Uncertainty. Uncertainty over acts of nature.

Definition 2. Strategic uncertainty. Uncertainty over actions taken by other rational agents.

Game theory is about strategic uncertainty.

1.2 Choice

Economics and game theory is about humans who have agency, which we call agents or players. The agents' behavior is to make choices over a preference relation. We say that $a \succsim b$ when˙ the agent chooses a from the set $\{a, b\}$.

Formally, let O be the set of possible outcomes. We let a binary relation $\succsim$ on O represent an agent's choices. We say that the preferences are rational if there is a utility function $U : O \to \mathbb{R}$ s.t. $a \succsim b$ iff $U(a) \geq U(b)$. If so, then we say that $\succsim$ is complete and transitive. Note that the choice behavior $\succsim$ is observable, but U is not.

1.3 Uncertainty

How do model behavior under uncertainty? There three ways to think of this problem.

1.3.1 Risk

The first is choice under **risk**. Agents must choose among actions a or b where the consequences of each actions depends on random outcomes that have **known probability distributions.**

A classic example of choice under risk is when agents must choose actions a or b where the consequence of each action depends on a coin toss with probability p and $1 - p$.

	H	T
a	$u(x)$	$u(y)$
b	$u(z)$	$u(w)$

The agent will decide to choose a from $\{a, b\}$, or $a \succsim b$ iff

$$\mathbb{E}_p u(a) \geq \mathbb{E}_p u(b) \Leftrightarrow pu(x) + (1-p)u(y) \geq pu(z) + (1-p)u(w)$$

In this case, we have that $\succsim$ and p are **observable**, but u may not be.

1.3.2 Subjective Expected Utility

The previous example modeled choices under risk, but how do we model agents choosing among **uncertain** bets? Again, agents must choose among actions a or b where the consequence of each action are unknown. There are two possible states of the world, ω_1 and ω_2 where the consequences of each action depend on the unknown state of the world.

In this version, agents behave as if they had subjective **beliefs** over the state of the world. Again, agents choose a from $\{a, b\}$ iff

$$pu(x) + (1-p)u(y) \geq pu(z) + (1-p)u(w)$$

where p is the probability of state ω_1 and $u(o)$ is the utility of outcome o for $o \in \{x, y, z, w\}$. In this version, $\succsim$ is observable, but u **and** p are not. This version of uncertainty operates under **Subjective Expected Utility**.

1.3.3 Strategic Uncertainty

The final version of uncertainty we will discuss is **strategic uncertainty**. In this version, the state of the world depends on other agents' choices, and the outcome depends on what other agents choose to do. Strategic uncertainty falls under the realm of game theory.

Suppose we have Agent B who chooses from $\{c, d\}$

	c	d
a	$u(x)$	$u(y)$
b	$u(z)$	$u(w)$

Again, Agent A will choose a or b iff

$$pu(x) + (1-p)u(y) \geq pu(z) + (1-p)u(w)$$

where p is the subjective belief that the second player chooses c.

1.4 Domination and Justifiability

How should players choose actions?

1.4.1 Actions and mixed actions

Let A be a finite set of possible **actions** and let Ω be a finite set of possible **states**. We say that each action $a \in A$ has a state-contingent payoff, or utility, $u(a, \omega)$ for each $\omega \in \Omega$. We can represent the utility of each action under states as a vector $\vec{u}(a) = (u(a, \omega))_{\omega \in \Omega} \in \mathbb{R}^{\Omega}$. We assume that the utilities/payoffs are given or known – this is common in game theory.

Let $\Delta(A)$ be the set of probability distributions over A. The elements $\sigma \in \Delta(A)$ are called **mixed actions**. We imagine agents who may choose an action at random, i.e., agents are choosing probability distributions over their actions.

Let $\Delta(\Omega)$ be the set of **beliefs** our agents may have. Given $\mu \in \Delta(\Omega)$, the expected utility from action a is

$$\mathbb{E}_{\mu} u(a, \tilde{\omega}) = \sum_{\omega \in \Omega} \mu(\omega) u(a, \omega)$$

Which gives us the expected utility from a given action over the distribution of beliefs. (Notation: $\tilde{x}$ implies we are integrating over object x) Thus, given a mixed action $\sigma \in \Delta(A)$, the expected utility is

$$\mathbb{E}_{\mu} u(\sigma, \tilde{\omega}) = \mathbb{E}_{\sigma} \mathbb{E}_{\mu} u(\tilde{a}, \tilde{\omega}) = \sum_{a \in A} \sigma(a) \mathbb{E}_{\mu} u(a, \tilde{\omega}) = \sum_{a \in A} \sigma(a) \sum_{\omega \in \Omega} \mu(\omega) u(a, \omega) = \mathbb{E}_{\mu} \mathbb{E}_{\sigma} u(\tilde{a}, \tilde{\omega}) = \mathbb{E}_{\sigma} u(\tilde{a}, \omega)$$

The above is basically an application of Fubini's theorem which lets you switch the summations. (Notation: if p a prob dist, then $f(p) = \mathbb{E}_p(f) = \int f dp$).

1.4.2 Optimality

A mixed action $\sigma \in \Delta(A)$ is **optimal** given beliefs $\mu \in \Delta(\Omega)$ if

$$\mathbb{E}_\mu u(\sigma, \tilde{\omega}) \geq \mathbb{E}_\mu u(\sigma', \tilde{\omega})$$

for all $\sigma' \in \Delta(A)$. We say that a mixed action is **justifiable** if it is optimal given **some** beliefs.

1.4.3 Domination

We can also make conclusions about optimality from actions independent of beliefs. A mixed action $\sigma \in \Delta(A)$ is **dominant** if

$$u(\sigma, \omega) > u(\sigma', \omega)$$

for all $\omega \in \Omega$, $\sigma' \in \Delta(A)$, $\sigma' \neq \sigma$.

A mixed action $\sigma \in \Delta(A)$ is **dominated** if there exists $\sigma' \in \Delta(A)$ such that

$$u(\sigma, \omega) < u(\sigma', \omega)$$

for all $\omega \in \Omega$. In this case, we say that σ' dominates σ. More precisely for the examples above, we say that σ' **strictly dominates** σ, but for now just dominates is fine.

1.4.4 Domination theorem

Theorem 3. *A mixed action is justifiable if and only if it is not dominated.*

To say that σ is not dominated seems like a weaker statement that to say that σ is a best response to some belief, but it's actually not weaker. One direction is trivial, but the other is not.

> Consider the following application to Statistical Decision Theory. A statistical decision problem is given by $((\Omega, P), A, L)$ in which Ω is a probability space, $P = (P_\omega)_{\omega \in \Omega}$ is a probability distribution for ω on the sample space X, A is a set of possible actions, and $L : \Omega \times A \to \mathbb{R}$ is a loss function. A common loss function is squared error loss, $L(\omega, a) = (\omega - a)^2$. A decision rule is a function $d : X \to A$ where often $A = \Omega$ and the decision is to estimate a parameter. The underlying ω is unknown but the statistician observes x and decides and estimate $d(x)$. A decision rule d can be evaluated according to its risk if the parameter is $\omega : R(\omega, d) = \int_X L(\omega, d(\tilde{x}))\, dP_\omega(\tilde{x})$. The domination theorem says that a decision rule is undominated (admissible) iff it is optimal for some prior distribution over parameters (this is the application of the domination theorem) Such decision rules are called Bayes optimal rules. This prior is what we called beliefs.

We will prove the domination theorem using the separating hyperplane theorem (SHT). Before getting into the proof, recall the SHT.

Theorem 4. *Separating Hyperplane Theorem. Let $A, B \subseteq \mathbb{R}^m$ be non-empty, disjoin, convex sets. There is $p \in \mathbb{R}^m$, and $\alpha \in \mathbb{R}$, such that*

$$p \cdot b \leq \alpha \leq p \cdot a$$

for all $a \in A$ and $b \in B$, and at least one of the inequalities is strict for some $a \in A$ and $b \in B$ and $p \neq 0$.

*Here the set $\{x : p \cdot x = \alpha\}$ is a **hyperplane**. It defines two half-spaces: $\{x : p \cdot x \leq \alpha\}$ and $\{x : p \cdot x \geq \alpha\}$. As a notational shortcut, we will also describe the SHT as $p \cdot B \leq \alpha \leq p \cdot A$.*

Proof. Proof of the Domination Theorem. Let σ^* be undominated. Define the set $\mathcal{U} = \{\vec{u}(\sigma) : \sigma \in \Delta(A)\}$ to be all possible mixes strategies which can be created from the pure strategies given the set of beliefs. Note that since σ^* is undominated, then there are no mixed strategies to the upper-right of this action. The set,

$\mathcal{U}$ is convex. Now define the set $\mathcal{C}$ to be the set of strategies that strictly dominate σ^*, $\mathcal{C} = \left\{\vec{u}\left(\sigma^*\right) + \mathbb{R}^{\Omega}_{++}\right\}$. This set is also convex. We can now apply the SHT to $\mathcal{U}$ and $\mathcal{C}$ and let (p, α) define the hyperplane with

$$p \cdot \mathcal{U} \leq \alpha \leq p \cdot \mathcal{C}$$

The interpretation here is that we can turn p into a belief, i.e., p is a probability distribution. For this to be true, the $p \geq 0$ and must sum to one.

Claim: $p \geq 0$. Suppose BWOC $p\left(\hat{\omega}\right) < 0$ for some $\hat{\omega}$. Pick any σ and for $x \in \mathcal{C}$ define

$$x^n\left(\omega\right) = \begin{cases} x\left(\omega\right) & \omega \neq \hat{\omega} \\ x\left(\omega\right) + n & \omega = \hat{\omega} \end{cases}$$

Then, $p \cdot \vec{u}\left(\sigma\right) \leq p \cdot x^n = p \cdot x + p\left(\hat{\omega}\right) n$ which cannot hold for arbitrarily large n since $p\left(\hat{\omega}\right) < 0$. Thus, $p \geq 0$ and by SHT, $p \neq 0$, so $p > 0$.

Define $\mu = \frac{1}{\sum_{\omega \in \Omega} p(\omega)} p$, and observe that $\mu \in \Delta\left(\Omega\right)$. Thus, p is a probability distribution. $\square$